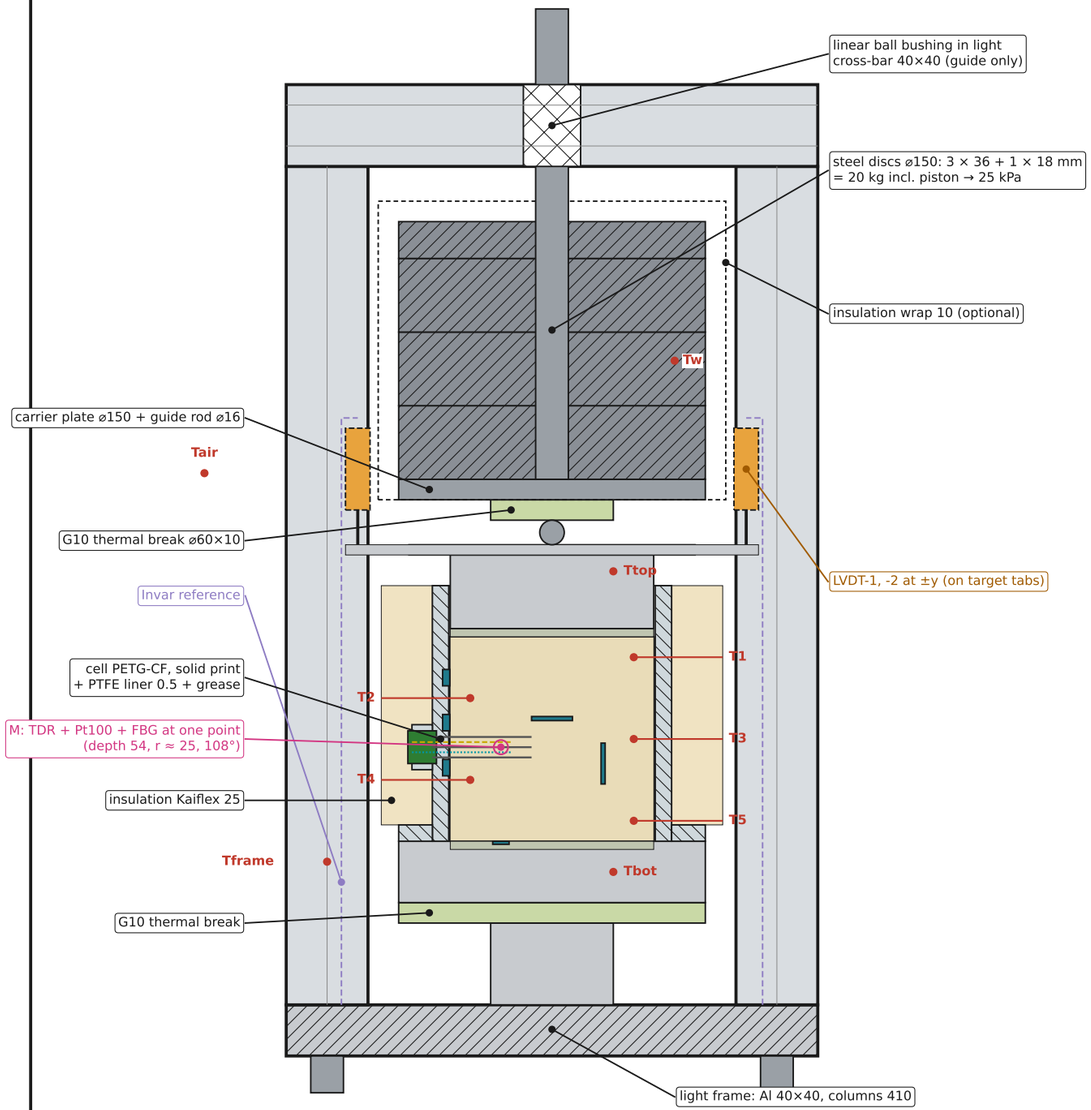


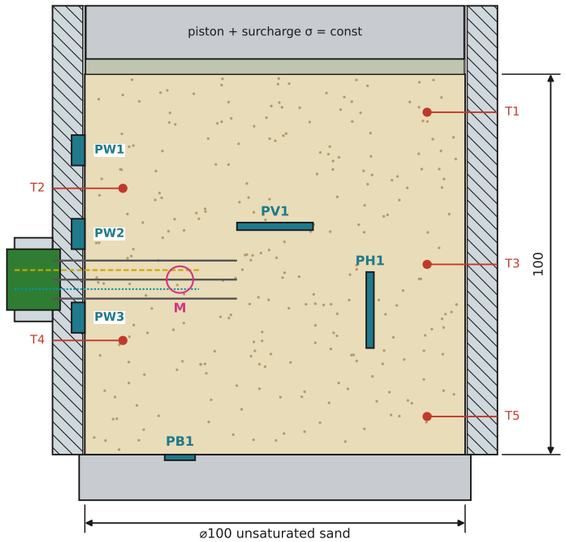
Sensors (tag → model → CR1000)

tag	measures	recommended model	CR1000
PV1 / PH1	vert. / radial stress	TML PDA-PA 200 kPa on ø20×2 disc	MUX-1 (4×16), BrFull, VX3 2 V
PB1	vertical stress at base	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
PW1…PW3	lateral stress on wall	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
LVDT-1,2	check: no movement	RDP DCTH400	SE1–2, VoltSE ±5 V
T1…T5	soil, r 40 mm	Campbell 109SS	MUX-2 → SE13, VX2
M: TDR	water / unfrozen water	mini 3-rod TDR 40 mm [U] + TDR200	SDM C1–C3
M: Pt100	reference temperature	Pt100 1/10 DIN, 4-wire, ø3	MUX-1 + 4WPB100, BrHalf4W
M: FBG	optical temperature	FBG temp. sensor in capillary ø1 [U]	PC (Ethernet), time-synced
Ttop, Tbot, Tw	piston, pedestal, weights	Campbell 109SS	(same)
Tframe, Tair	frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

σ kPa	total kg	full discs	half discs	trim kg
5	4.0	0	0	1.25
10	8.0	1	0	0.33
15	12.0	1	1	1.88
20	16.0	2	1	0.96
25	20.0	3	1	0.04

How it works
Gravity gives an exactly constant surcharge; fixed part ≈ 2.8 kg ≈ 3.4 kPa (piston, carrier, rod).
No load cell, no adjustment; the frame only guides the rod and carries the LVDTs.
Weights lag the freezer air by ≈ 0.6 K at the 0.1 K/h ramp (vypocty.md §7).

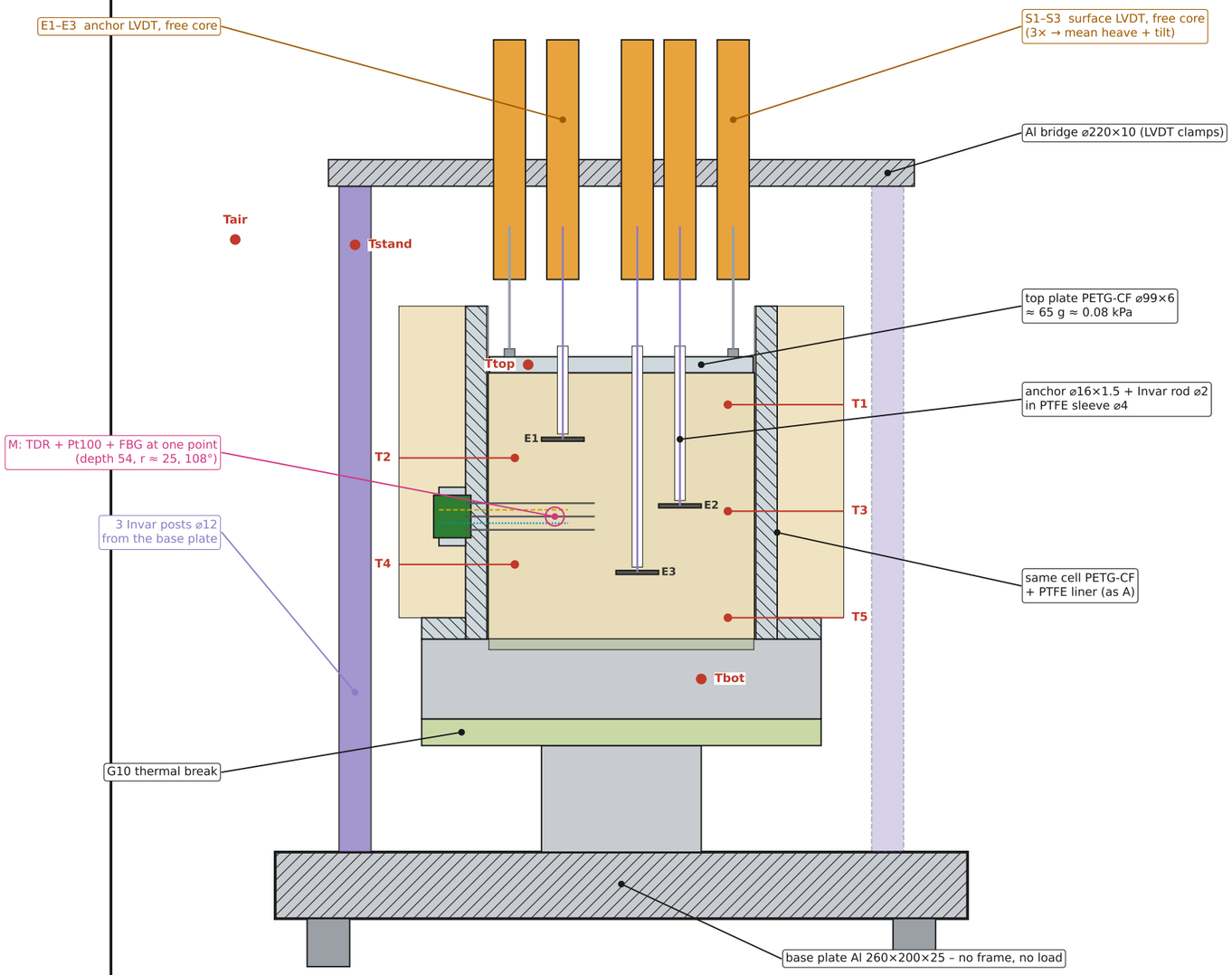
DETAIL – sand, pressure sensors, thermistors ≈ 1 : 1.6



schematic; orange = displacement, blue = pressure, red = temperature; cells and LVDTs revolved into section

Variant A – unsaturated sand, dead-weight surcharge ≤ 25 kPa

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 3**	drawing no.	**E7-V01**	sheet 1 / 7



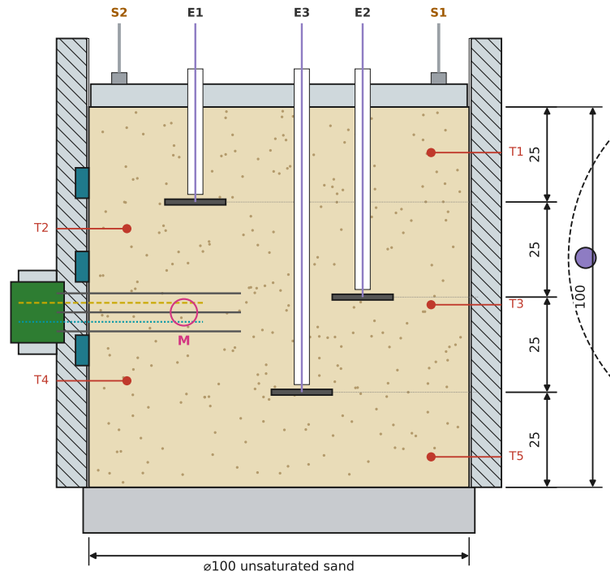
Sensors (tag → model → CR1000)

tag	measures	recommended model	CR1000
S1…S3	surface heave / settlement	RDP DCTH200, free core [U]	SE3–5
E1…E3	anchor displ. at 25/50/75	RDP DCTH200, free core [U]	SE6–8
PW1…PW3	lateral stress (optional)	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
T1…T5	soil, r 40 mm	Campbell 109SS	MUX-2 → SE13, VX2
M: TDR	water / unfrozen water	mini 3-rod TDR 40 mm [U] + TDR200	SDM C1–C3
M: Pt100	reference temperature	Pt100 1/10 DIN, 4-wire, ø3	MUX-1 + 4WPB100, BrHalf4W
M: FBG	optical temperature	FBG temp. sensor in capillary ø1 [U]	PC (Ethernet), time-synced
Ttop, Tbot	top plate, pedestal	Campbell 109SS	(same)
Tstand, Tair	Invar stand, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

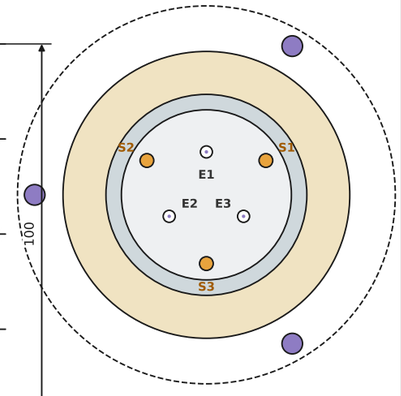
What is measured in each freeze–thaw cycle
- surface: heave during freezing, settlement during thawing, residual after the cycle (mean of S1–S3; their difference = tilt of the top plate)
- 4 layers of 25 mm: $\Delta h_1 = \bar{S} - E1$, $\Delta h_2 = E1 - E2$, $\Delta h_3 = E2 - E3$, $\Delta h_4 = E3$ (base fixed) strain $\varepsilon_i = \Delta h_i / 25$ mm, resolution ≈ $4 \cdot 10^{-5}$ (vypocty.md §3)
- frost depth from T1–T5 (0 °C isotherm) → strain vs. frozen / unfrozen layer
- after the last cycle: height (caliper), mass, water content of 5 slices × 20 mm

No load: free-core LVDTs push nothing; the plate adds 0.08 kPa only.
Reference error ≈ 50 µm per 20 K → blank run with Al dummy (vypocty.md §4).

DETAIL – anchors and top plate ≈ 1 : 1.6



PLAN – top plate 1 : 3.6

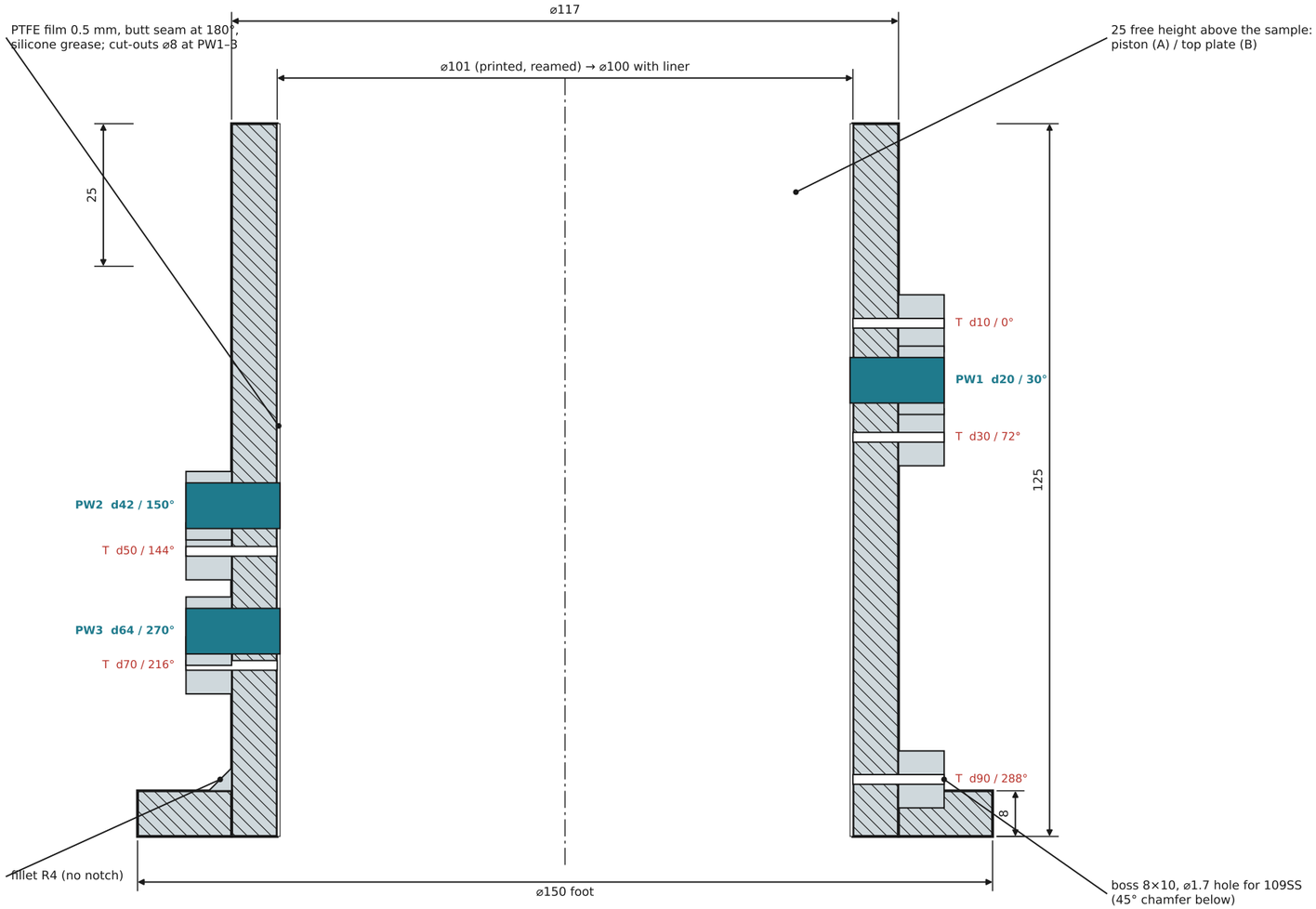


orange S = surface LVDT (r 40)
E = anchor rod hole (r 25)
violet = Invar posts (r 100)

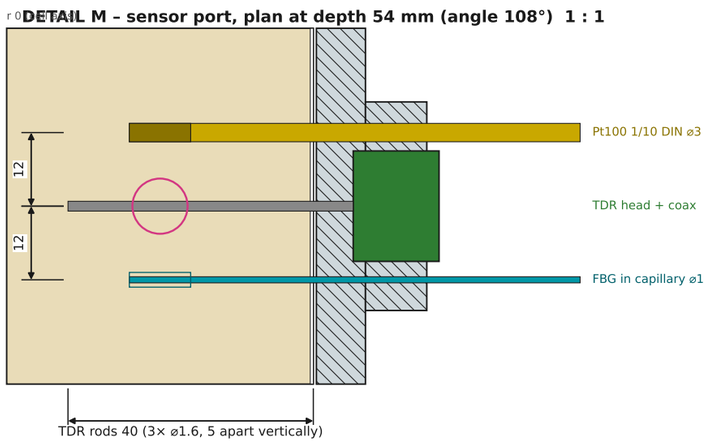
schematic; orange = displacement, red = temperature, violet = Invar reference; sensors revolved into section (true positions on the plan)

Variant B – same sample, no load: heave, settlement, layer strain

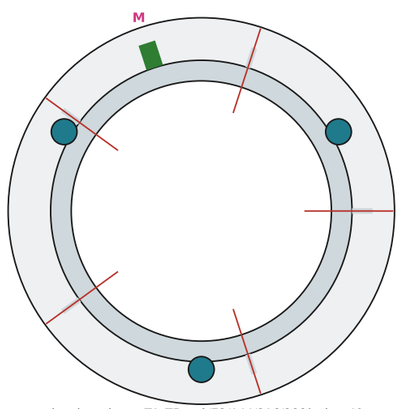
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2**	drawing no.	**E7-V02**	sheet 2 / 7



VIEW F – cell from above 1 : 2.4



DETAIL M – sensor port, plan at depth 54 mm (angle 108°) 1 : 1



red = thermistors T1–T5 at 0/72/144/216/288°, tip r 40;
blue = PW1–3 at 30/150/270°; green = point M at 108°

Material check – PETG-CF

load case	stress	SF short	SF long
hoop: A 25 kPa surcharge, K0 = 1	0.16 MPa	> 250	> 75
hoop: B ice on the wall 200 kPa (upper est.)	1.3 MPa	32–44	9–13
hoop: extreme 1 MPa	6.3 MPa	6–9	1.9–2.6
axial (across layers): friction, A 25 kPa	< 0.05 MPa	> 300	> 90

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §5.

Verdict: suitable for both variants as a solid print (no sleeve needed).

Wall friction (the price of a solid cell, vypocty.md §5)
- A: friction carries part of the surcharge into the wall; with the PTFE liner + grease (μ ≈ 0.05) the vertical stress at the base is ≈ 90 % of the top, without the liner (μ ≈ 0.5) only ≈ 30–45 % → compare PV1 / PB1 with it.
- B: friction and ice adhesion hold back heave of the frozen top layer; PTFE has very low ice adhesion – liner and grease are mandatory.

Conditions
- wall 8 mm, concentric perimeters only, 100 % infill;
- no printed threads or sharp notches (CF lowers toughness at −10 °C);
- inside sealed with a thin epoxy coat before the liner goes in;
- cell not clamped at the foot – friction then pushes, never pulls layers.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.
Print upright, bosses with 45° chamfer underneath (no supports).

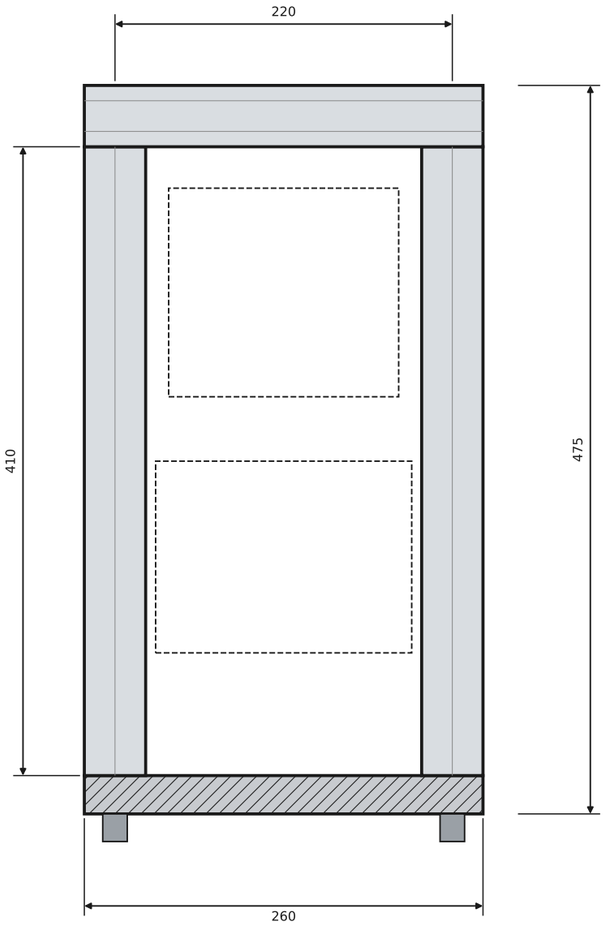
Qualification before use
1 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
2 bore check ø101 ± 0.1 at 3 heights, at +20 and −10 °C.

print upright in one piece; hoop direction = layer direction; foot rests on the pedestal, not bolted

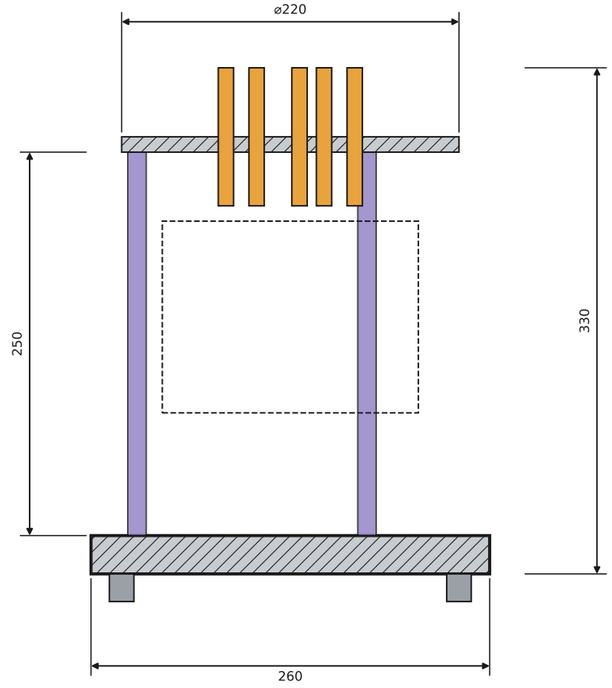
Sample cell – PETG-CF, solid 3D print (both variants)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1**	drawing no.	**E7-V03**	sheet 3 / 7

A – FRONT VIEW 1 : 4



B – FRONT VIEW 1 : 4



Bill of materials

#	part	size	qty
1	base plate Al 6082 T6 (A and B)	260 × 200 × 25	2
2	A: profile 40×40 column	l = 410	2
3	A: profile 40×40 cross-bar	l = 260	1
4	A: linear ball bushing ø16 in cross-bar		1
5	A: Invar post ø10 + clamp arm (LVDT)	l ≈ 300	2
6	B: Invar post ø12, M6 both ends	l = 250	3
7	B: Al bridge ø220×10, 6 LVDT clamps		1
8	core bolt M8 / socket screw M8		8
9	levelling foot M12, steel		6

Frame A: carries nothing except the guide of the weight rod (bushing) and the LVDT posts; the surcharge goes straight down through the sample.

Stand B: three Invar posts screwed into the base plate at r 100 (60/180/300°, outside the insulation), bridge on top with split clamps for 5 LVDTs (+1 spare). The bridge is lifted off to fill the cell. Reference error of the stand ≈ 50 µm per 20 K (Al pedestal vs. Invar) → blank run with an Al dummy sample (vypocty.md §4).

Heights incl. base plate: A 475 mm (+ guide rod), B 330 mm; two rigs side by side on the tray – layout on E7-V06.

frame A only guides the weights; stand B is only a displacement reference – neither carries load

Frame A (aluminium profiles) and Invar stand B

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 4**	drawing no.	**E7-V04**	sheet 4 / 7

Options per measurement (PICK = recommended)

sensor	maker / model	key specs	verdict
LVDT ±10	RDP DCTH400	±5 V / 0–10 V out, −40…+80 °C, ±0.25 % FS	PICK (A)
LVDT ±10	Micro-Epsilon DTA-10D + MSC7401	0–5 V / 4–20 mA, −20…+80 °C	alternative
LVDT ±10	Solartron AX/10/S + OD5	±5 V out, needs 230 V conditioner	alternative
LVDT ±10	HBM WA-L 20 mm	passive, needs carrier amplifier	alternative
LVDT free	RDP DCTH200 unguided [U]	±5 mm, no spring, core on ø3 rod; rig B	PICK (B)
LVDT free	Micro-Epsilon DTA-5G8 + MSC7401 [U]	±5 mm, free core, 0–5 V, −20…+80 °C	alternative
LVDT free	Solartron AX/5/F (free core) + OD5 [U]	±5 mm, ±5 V via conditioner	alternative
laser	Micro-Epsilon optoNCDT 1420	non-contact, but op. 0…+50 °C, frost on window	not suitable
LVDT ±10	TE DC-EC 500	operating only 0…+70 °C	not suitable
p cell	TML PDA-PA 200 kPa	ø6.5×1 mm, 0.8 mV/V, comp. −10…+60, op. −20…+70 °C	PICK
p cell	Kyowa BEC-A 200/500 kPa	ø30 mm, for model tests; t, temp. [U]	one cell only
p cell	TML KDE/KDF-PA	ø50 mm – too large for ø100 sample	not suitable
p wall	TE/MEAS EPL	miniature, comp. −40…+90 °C, ø [U]	alternative
p cell	Tekscan FlexiForce A201	thin film, drift < 5 %/log t	comparison only
T probe	Campbell 109SS	ø1.6 mm, ±0.38 °C at 0 °C, direct on CR1000	PICK
T probe	Campbell 107	ø7.6 mm, ±0.2 °C (0–50 °C)	air / frame
T probe	TE/YSI 44031 bead	±0.1 °C, own half bridge needed	for 0.01 °C (E6 chain)
Pt100	Pt100 1/10 DIN 4-wire ø3 + 4WPB100	±0.03 °C at 0 °C, via MUX-1	PICK (point M)
TDR	mini 3-rod 40 mm (custom) + TDR200	fits ø100 sample; calibrate in the sand	PICK [U]
TDR	Campbell CS645 (3 rods 75 mm)	too long for r 50 – would cross the axis	not suitable
TDR	Easy Test LP/ms micro probe + own meter [U]	rods ≈ 53 mm; separate logger	alternative
optical	FBG in capillary ø1 + HBM FS22 / Luna si155 [U]	≈ ±0.2–0.5 °C, many points on one fibre	PICK [U]
optical	fluorescent probe Opsens OTG-F + TempSens [U]	±0.3 °C, analog out → CR1000 SE	alternative

CR1000 channel plan

terminal	connected to	measurement
SE1–SE2	LVDT-A1, A2 (rig A)	VoltSE ±5000 mV, averaging; own ±15 V supply
SE3–SE5	LVDT S1–S3 (rig B surface)	VoltSE ±5000 mV
SE6–SE8	LVDT E1–E3 (rig B anchors)	VoltSE ±5000 mV
DIFF5, DIFF6	MUX-1 AM16/32B, 4×16	9 p-bridges + 2 Pt100 (4WPB100); COM H2/L2 → DIFF5
VX3	MUX-1 excitation	2000 mV bridges / 2500 mV Pt100
C1–C3	TDR200 (SDM) → SDMX50SP	2 TDR probes (point M, A and B)
C4, C5	MUX-1 reset, clock	
SE13	MUX-2 AM16/32B, 2×32	≈ 20 thermistors (Therm109), VX2
C6, C7	MUX-2 reset, clock	
12V, G	both multiplexers, TDR200	
–	FBG interrogator	1 fibre for both rigs; own PC, clock synced
SE14–16	spare	

Channel count (2 rigs): 8 LVDTs, 9 pressure bridges, ≈ 20 thermistors,
point M ×2: 2 TDR (TDR200 + SDMX50SP), 2 Pt100 (MUX-1), 2 FBG (interrogator).
No load cells: A is loaded by dead weight, B is not loaded at all.

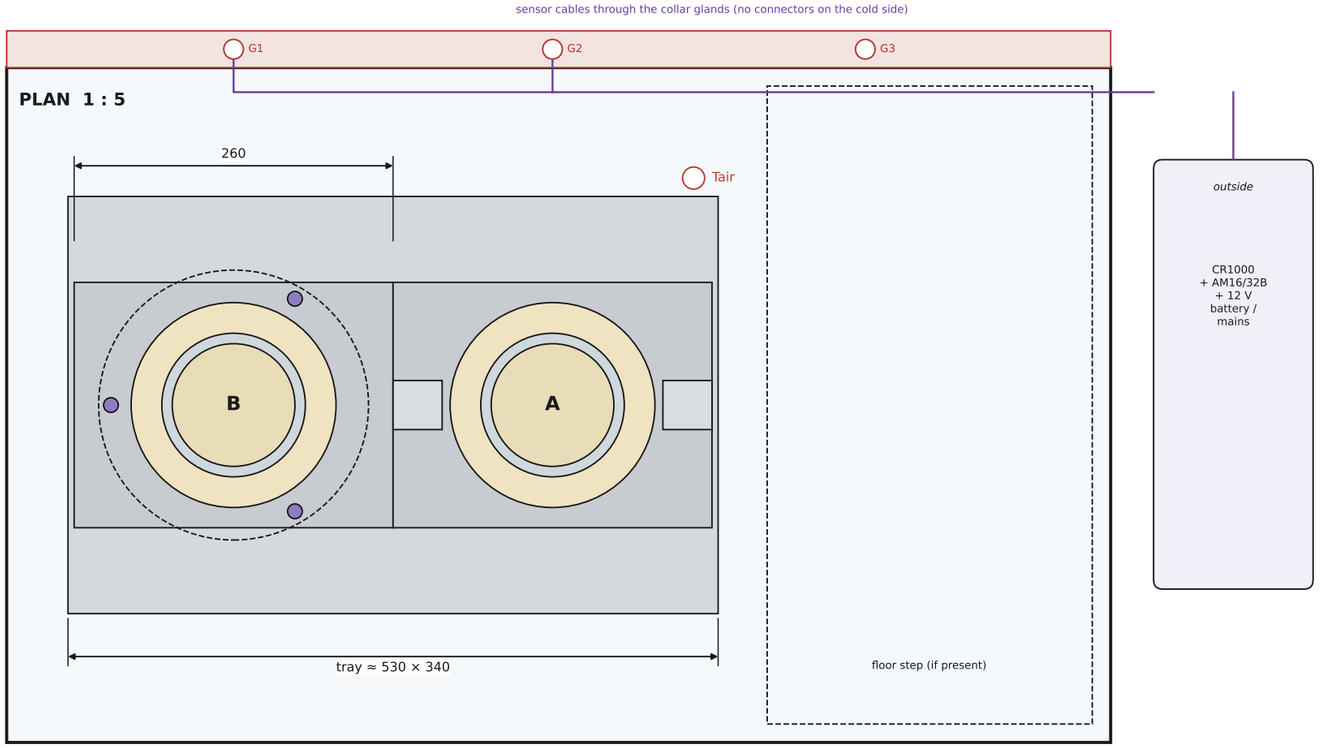
Resolution with the CR1000 (13-bit ADC, ±5000 mV range)
- LVDT ±10 mm (A): ≈ 1.3 µm; LVDT ±5 mm (B): ≈ 0.7 µm
  → layer strain over 25 mm ≈ $4 \cdot 10^{-5}$ (B)
- absolute linearity ±0.25 % FS → calibrate each LVDT with a micrometer
- 109SS: ±0.38 °C at 0 °C; ice-point offset check improves it

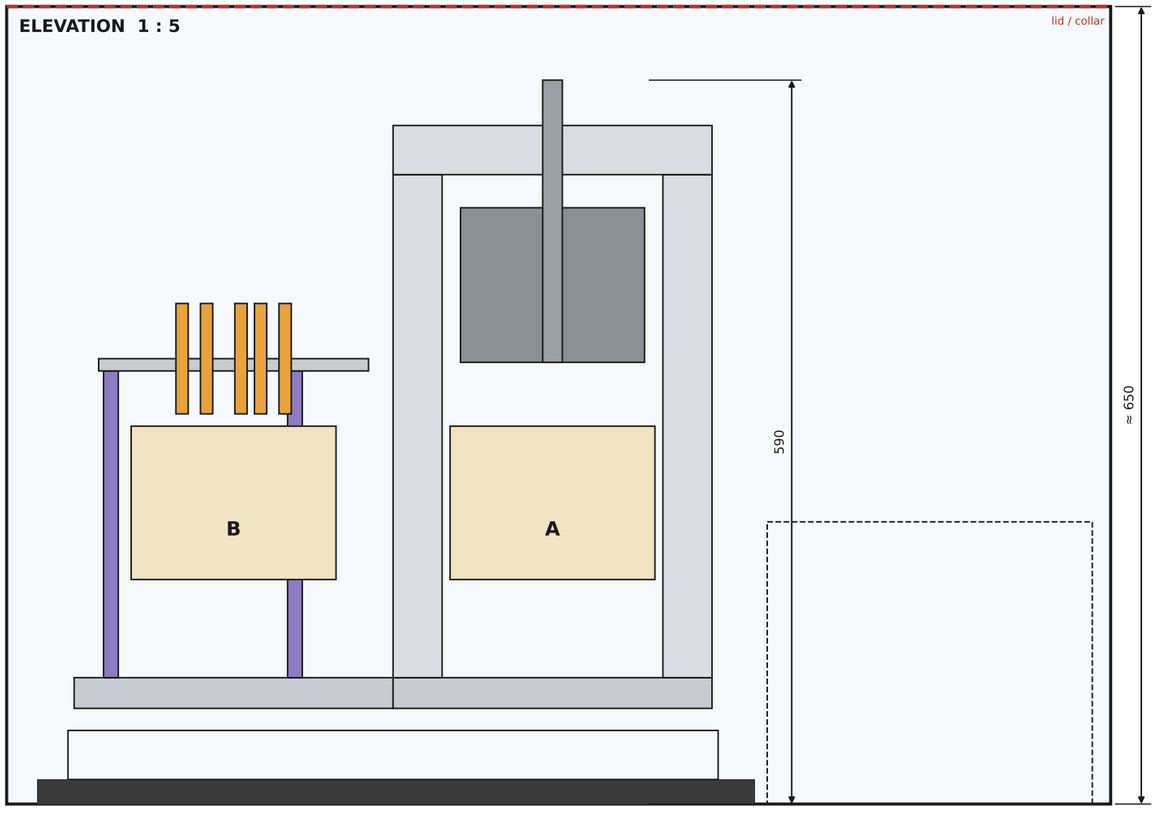
Notes
- Rig B uses LVDTs with a FREE (unguided) core – no spring force on the sample; the core rides on the anchor rod / plate rod. Check the free-core version and the core length with RDP before ordering [U].
- DC-LVDTs need their own ±15 V (or 24 V) supply, logger only reads ±5 V.
- CR1000 is discontinued (successor CR1000X / CR1000Xe, 24-bit); the existing one is fine for these resolutions.
- Logger outside the freezer (CR1000 −25…+50 °C).
- Point M: TDR, Pt100 and FBG sit 12 mm apart at one depth – Pt100 is the reference for the FBG and for T3; TDR gives unfrozen water vs. temperature.
- Pressure cells: calibrate in the same sand under dead weight (5–25 kPa) at +10 and −10 °C; correct zero(T) with the neighbouring thermistor.

both rigs on one CR1000 + AM16/32B; models from manufacturer data (see README §7)

Sensor selection and CR1000 wiring

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	–	drawing no.	**E7-V05**	sheet 5 / 7





Fit check

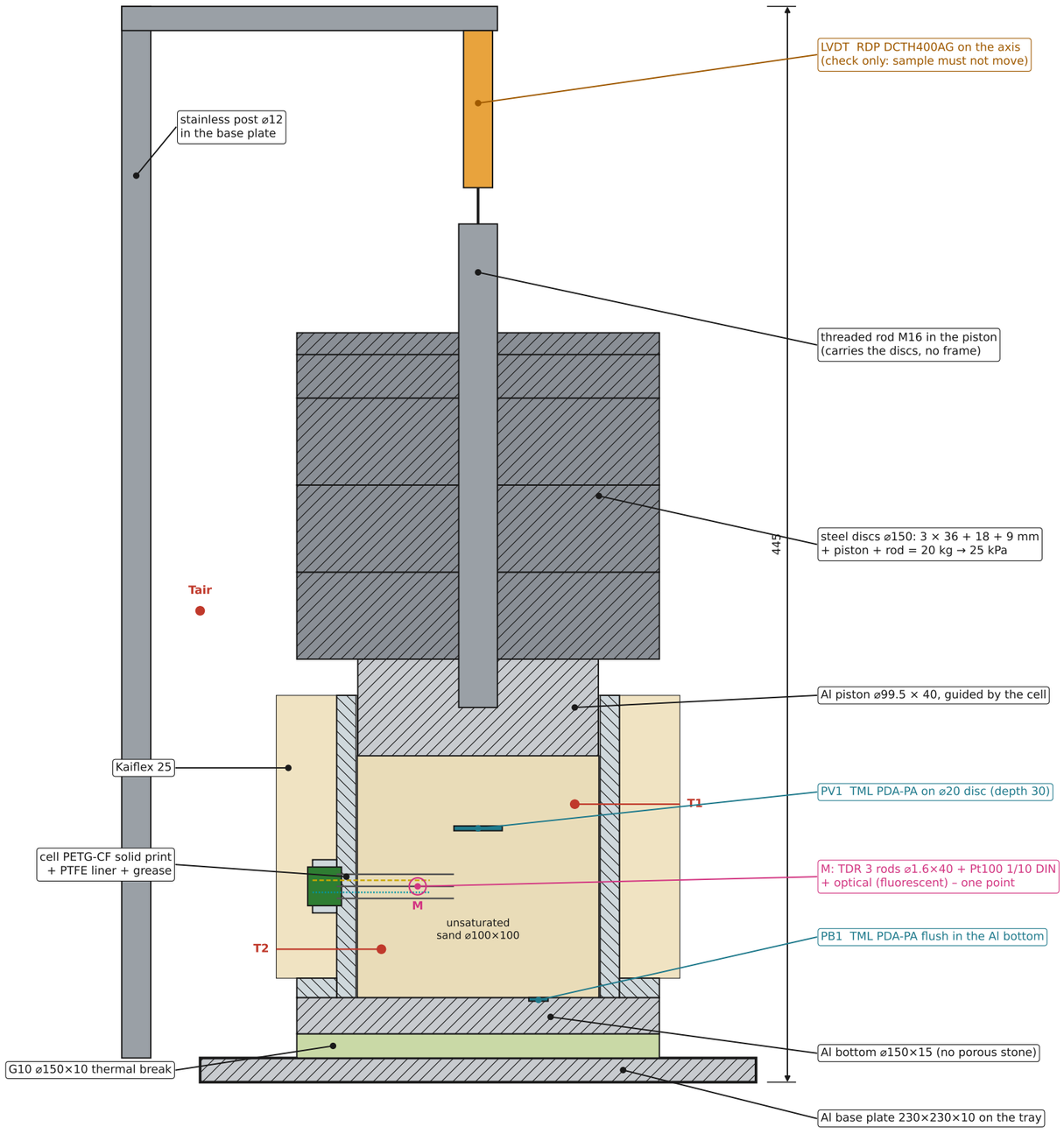
item	size	limit	status
base plates 2 × 260 × 200	520 × 200	tray ≈ 530 × 340	OK (tight)
rig A (weights)	20 kg steel discs ø150	nothing outside frame	OK
rig B (stand)	408 mm high	no load, no frame	OK
highest point	590 mm	inside ≈ 650	OK, measure
logger	outside, 0…+40 °C	CR1000 −25…+50 °C	keep warm & dry

Notes
- Both rigs share the freezer program (Set A) and the CR1000; each rig has its own Tair sensor at sample mid-height.
- Same sand, same printed cell, same thermistors in A and B – the only difference is the load (A ≤ 25 kPa dead weight, B none).
- Run the blank test with Al dummies in both rigs before the first sand run.
- Dimensions marked ≈ are estimates from the photos (E6) – please measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); logger outside

Both rigs in the chest freezer + CR1000

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V06**	sheet 6 / 7



Sensors – type, quantity, price (EUR excl. VAT, ≈)

tag	type	pcs	EUR	CR1000
PV1, PB1	TML PDA-PA 200 kPa (ø6.5 × 1 mm)	2	1 200–2 000	DIFF1, DIFF2 BrFull, VX1
Pt100	Omega PM, Pt100 1/10 DIN, 4-wire, ø3.2 × 150 mm	1	230–260 *	DIFF3+4 BrHalf4W, VX2
+	Campbell 4WPB100	1	60–120	–
TDR	Campbell TDR200 (reflectometer)	1	4 500–5 200 *	C1–C3 (SDM), 12V
+	TDR probe, custom 3 rods ø1.6 × 40 + RG58 1 m	1	50–150	–
OPT	Opsens OTG-F (fluorescent, ø0.15 mm)	1	300–600	–
+	Opsens TempSens 4-ch., ±5 V output	1	3 500–6 000	SE10 VoltSE
LVDT	RDP DCTH400AG (±10 mm, spring)	1	581 *	SE9 VoltSE ±5 V
+	DIN-rail power supply for the LVDT	1	40–80	–
T1, T2, Tair	Campbell 109SS-L (ø1.6 mm)	3	540–780	SE11–13 Therm109, VX3

sensors ≈ 11 001–15 771 EUR mechanics ≈ 380–780 EUR → total ≈ 11 381–16 551 EUR
* price found at a dealer (2026 see README); others = estimates prices are not published → ask for quotes.
CR1000 already available – not counted. TDR200 + optical converter ≈ 75 % of the sensor cost.

CR1000 wiring – no multiplexer
DIFF1 PV1, DIFF2 PB1 (BrFull, VX1 2000 mV) · DIFF3+4 Pt100 via 4WPB100 (BrHalf4W, VX2)
SE9 LVDT (±5 V) · SE10 optical converter (±5 V) · SE11–13 T1, T2, Tair (Therm109, VX3)
C1–C3 TDR200 (SDM), 12 V · SE14–16 spare

Removed compared with variant A (E7-V01)
• frame (2 columns + cross-bar + bushing), carrier plate, guide rod, ball, target plate
• G10 above the piston, porous stones, Al pedestal (→ one Al bottom disc)
• Invar posts, 2nd LVDT; pressure cells 6 → 2 (PV1, PB1); thermistors 10 → 3
• both AM16/32B multiplexers
Kept: same sand and cell, dead weight 5–25 kPa, point M (TDR + Pt100 + optical).
No frame: the piston is guided by the cell – tilt ≤ 1.1°, extra friction ≈ 0.4 kPa
(vypocty.md §9). Put the discs on slowly and centred.

schematic; blue = pressure, red = temperature, magenta = point M (TDR + Pt100 + optical); sensors revolved into the section

Variant A-min – simplified: dead weight ≤ 25 kPa, no frame, no multiplexer

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2.2**	drawing no.	**E7-V07**	sheet 7 / 7